LLL Failures on the Boneh–Venkatesan Lattice Were Floating-Point Artifacts: A Corrected Note

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Abstract

An earlier version of this note (PDF of 2026-10-11) reported two LLL failure modes for the Boneh–Venkatesan Hidden-Number-Problem lattice on deployed ECDSA curves: a "curve-specific" failure for secp256k1 at 256 bits and a uniform failure at ≥ 384 bits attributed to an undersized iteration cap. **Both were bugs in our floating-point Gram–Schmidt orthogonalisation, not properties of the curves or of LLL.** For secp256k1, and for every 384-bit curve, the squared Gram–Schmidt norms overflow IEEE double precision ($n^4 >$ `f64::MAX`). P-521 still fails on the scaled `f64` path, for a reason we have not isolated. Both fixes (scaling, and a 2048-bit fixed-point Gram–Schmidt) were in the companion code before that version was built. With them, a re-run on 2026-10-16 recovers the key for every distinct seed pair at 256 and 384 bits, secp256k1 included (3 of 3 distinct seed pairs, 6 probe runs), and on P-521 with 2048-bit fixed-point Gram–Schmidt (3 of 3; at $k_{\text{bits}} = 384$). We withdraw the curve-specificity claim and every conclusion drawn from it. What remains is a numerical lesson: an `f64` Gram–Schmidt step silently fails once the group order passes about 2^{256} (basis entries about 2^{512}).

Correction (2026-10-16)

This version replaces the note of 2026-10-11. An integrity audit of our results (the `sheriff` case `SHF-20261016-3caa6f`, finding P1, in the companion research repository) found that the note reported as curve properties two failures that our own analysis (`research/notes/lattice-hnp/RESEARCH_LLL_GS_ANALYSIS.md`, §10) had traced to numerical bugs and fixed in May 2026. The following claims of the earlier version are **withdrawn**:

- that secp256k1's LLL failure is "a curve-specific arithmetic phenomenon", "empirically robust" and "reproducible";
- that the ≥ 384-bit failures were caused by the LLL iteration cap;
- that LLL-convergence benchmarks are curve-specific and that reported HNP attack costs do not transfer between curves;
- the proposed curve-selection criterion against near-power-of-two group orders, and the open questions built on the curve-specificity claim.

The pre-fix measurements are kept below (Section 2) as the record of what the buggy code did.

Correction (2026-10-17)

A re-audit (`sheriff` case `SHF-20261017-e4d7f6`, findings P-N1 and P-N2) found two defects in the 2026-10-16 version. (i) It asserted a mechanism for the P-521 failure on the scaled `f64` path: "catastrophic cancellation", in which orthogonalising a row "subtracts nearly equal 53-bit quantities, and the residual swamps the true component". That mechanism is unsupported. With our scaling, the P-521 diagonal $n^2/2^{542}$ rounds to exactly 2^{500} in `f64`, so the subtraction described is exact on the initial basis[1], and the residual-dominates condition also holds for P-256, brainpoolP256r1 and brainpoolP384r1, which recover.

[1] Correction (2026-10-28, `sheriff` case `SHF-20261028-0f3f64`, PPN-2): this holds because the conversion is correctly rounded. The locked `num-bigint` 0.4.8 `to_f64` keeps the top 64 bits with a round-to-odd sticky bit and then rounds to nearest. Our scaled conversion first truncates 22 low bits of the 1042-bit n^2, far below the 53-bit significand. A truncating BigInt-to-`f64` conversion would instead give $2^{500}(1 - 2^{-53})$ and a residual of 2^{447}. This was checked by reading the source, not by a committed run.

The sentence is withdrawn; the measured result (scaled `f64` 0 of 3 distinct seed pairs at P-521, 2048-bit fixed point 3 of 3) stands. (ii) Table 2 counted probe runs of deterministic, overlapping test functions as if they were distinct instances, and said $k_{\text{bits}} = 3n_{\text{bits}}/4$ throughout, whereas every P-521 probe ran at $k_{\text{bits}} = 384$ (3/4 of 512, not of 521). The table now gives distinct seed pairs, runs and k_{bits}.

1 Setting and notation

The Boneh–Venkatesan lattice attack on biased-nonce ECDSA [1, 2] recovers the secret key d from m signatures (r_i, s_i, z_i) in which the nonce k_i is known to satisfy $0 \le k_i < 2^{k_{\text{bits}}}$ for some $k_{\text{bits}} < n_{\text{bits}}$. With $a_i = s_i^{-1} r_i \bmod n$ and $t_i = s_i^{-1} z_i \bmod n$, each signature contributes the constraint

$$d \cdot a_i + t_i \equiv k_i \pmod{n}.$$

The standard $(m+2)$-dimensional integer lattice basis (rows)

$$B = \begin{pmatrix} n^2 \cdot I_m & 0_{m\times 2} & \\ n \cdot \mathbf{a}^\top & 2^{k_{\text{bits}}} & 0 \\ n \cdot \mathbf{t}^\top & 0 & n \cdot 2^{k_{\text{bits}}} \end{pmatrix}$$

is fed to LLL with Lovász parameter $\delta = 0.75$; the short vector after reduction contains $-d \cdot 2^{k_{\text{bits}}}$ in the $(m+1)$-st coordinate (modulo sign).

2 What the earlier version measured

These are the pre-fix measurements as the 2026-10-11 version reported them. They were produced by unscaled `f64` Gram–Schmidt and are **superseded** by Section 4.

$(d_{\text{seed}}, k_{\text{seed}})$	P-256 outcome	P-256 time	secp256k1 outcome
`(0xC0FFEE, 0xC0FFEE)`	recovered	690 ms	iteration cap @ 9993 ms
`(0xDEADBEEF, 0xBADCAFE)`	recovered	697 ms	iteration cap @ 9845 ms
`(0x12345678, 0x9ABCDEF0)`	recovered	646 ms	iteration cap @ 10119 ms

Table 1: Pre-fix (superseded): $k_{\text{bits}} = 192$, $m = 8$, $\delta = 0.75$. Reported then as P-256 3/3, secp256k1 0/3.

The pre-fix sweep at $k_{\text{bits}} = 3n_{\text{bits}}/4$, $m = 8$, one seed, reported secp192k1, secp224k1, P-256 and brainpoolP256r1 recovering, and secp256k1, P-384, brainpoolP384r1 and P-521 hitting the iteration cap.

3 The causes

The basis has rows of size about n^2, so the squared Gram–Schmidt norms the reduction compares are of size about n^4.

Failure A: overflow. IEEE double precision tops out just below 2^{1024}. Among the three 256-bit curves, only secp256k1's order is close enough to 2^{256} that n^4 exceeds `f64::MAX`; P-256's and brainpoolP256r1's orders are smaller, and their n^4 stays below it (an exact rational comparison of the standard orders [3, 4], recorded in the audit as `checks/f64-overflow.txt`). The overflowed norms became infinite and then `NaN`, and LLL ran to its iteration cap. The "curve specificity" was the position of one group order relative to a floating-point limit. secp192k1 and secp224k1 escaped because their n^4 is far below the limit, which is also why the earlier near-power-of-two hypothesis seemed refuted. Scaling the basis by $2^{-(\text{max_bits}-500)}$ before orthogonalising removes the overflow (`src/cryptanalysis/lattice.rs`).

Failure B: overflow, and an unexplained P-521 failure. At 384 bits, n^4 overflows for every curve, which explains the "uniform" failure there; the same scaling fixes P-384 and brainpoolP384r1, so the iteration cap was not the cause. P-521 still fails after scaling (0 of 3 distinct seed pairs), and a 2048-bit fixed-point Gram–Schmidt (`lll_reduce_hp`) recovers it (3 of 3). We have not isolated where the scaled `f64` path loses precision at P-521; the cancellation argument of §10.2 of the analysis does not apply to the initial basis (correction of 2026-10-17 above), and locating the failing step would need the Gram–Schmidt instrumented during the probe.

4 Re-run with the fixes (2026-10-16)

We re-ran the committed probes (`tests/lll_degeneracy_probe.rs`) on the fixed code. Commands, logs, parsed results and the host are in `research/lll_degeneracy_rerun_20261016/`.

Curve	Gram–Schmidt	k_{bits}	Recovered / distinct	Runs
secp192k1	scaled `f64`	144	2 / 2	3 / 3
secp224k1	scaled `f64`	168	2 / 2	3 / 3
secp256k1	scaled `f64`	192	3 / 3	6 / 6
P-256	scaled `f64`	192	3 / 3	6 / 6
brainpoolP256r1	scaled `f64`	192	1 / 1	1 / 1
P-384	scaled `f64`	288	3 / 3	4 / 4
brainpoolP384r1	scaled `f64`	288	3 / 3	4 / 4
P-521	scaled `f64`	384	0 / 3	0 / 4
P-521	2048-bit fixed point	384	3 / 3	3 / 3

Table 2: Re-run, $m = 8$, $\delta = 0.75$. "Distinct" counts distinct $(d_{\text{seed}}, k_{\text{seed}})$ pairs; the probes are deterministic, and overlapping test functions repeat pairs, so "Runs" counts probe executions (the 2026-10-16 version reported runs only). $k_{\text{bits}} = 3n_{\text{bits}}/4$ except at P-521, where the probes use $384 = 3/4 \cdot 512$, not $\lfloor 3 \cdot 521/4 \rfloor = 390$. Each recovered probe returned the planted key. Source: `logs/probes.log` and `logs/p521_hp.log` in the re-run directory (corrected 2026-10-17).

Every probe at 256 and 384 bits finished in 13–27 ms on an unisolated 4-vCPU virtual machine, against the roughly 10 s and 20 s cap hits of the earlier version. These timings are single runs, not a benchmark.

5 What remains

No curve-specific LLL behaviour remains at these parameters. The lesson is about implementations: an `f64` Gram–Schmidt step fails silently, by overflow, once the squared norms pass 2^{1024}, that is once n passes about 2^{256} and the basis entries about 2^{512}. At P-521 the scaled `f64` path fails for a reason we have not isolated, while 2048-bit fixed point succeeds. An HNP experiment that reports an LLL failure should state its arithmetic and rule this out before attributing the failure to the curve.

This note makes no security claim about any curve. It reports a small number of probes, one to three distinct seed pairs per curve, at one parameter set ($m = 8$, $k_{\text{bits}} = 3n_{\text{bits}}/4$, except $k_{\text{bits}} = 384$ at P-521).

References

[1] D. Boneh, R. Venkatesan, "Hardness of computing the most significant bits of secret keys in Diffie–Hellman and related schemes", CRYPTO 1996.

[2] P. Nguyen, I. Shparlinski, "The insecurity of the digital signature algorithm with partially known nonces", *J. Cryptol.* 2002.

[3] NIST FIPS 186-4, "Digital Signature Standard", 2013.

[4] Certicom Research, "SEC 2: Recommended Elliptic Curve Domain Parameters", 2010.